

Automated Curriculum Vitae Parsing and Candidate Registration System Using n8n Workflow Automation, Google Gemini Artificial Intelligence, and Notion Database Integration

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Abstract—This paper presents the design, implementation, and empirical validation of a fully automated Curriculum Vitae (CV) parsing and candidate registration system. The proposed system integrates the n8n open-source workflow automation platform with the Google Drive API, the Google Gemini 1.5 Flash large language model (LLM) API, and the Notion database API to construct an end-to-end, event-driven processing pipeline. Upon the upload of a candidate CV in Portable Document Format (PDF) to a designated Google Drive folder, the workflow is triggered automatically, the document is retrieved and its textual content extracted, and the unstructured text is then submitted to the Gemini 1.5 Flash model for intelligent parsing. The model extracts structured candidate profile information—comprising the full name, electronic mail address, telephone number, professional designation, and a narrative summary—and the resulting structured data record is subsequently persisted as a new entry in a relational Notion database. The system eliminates manual data entry entirely, reduces processing latency to a matter of seconds, and demonstrates the practical viability of combining low-code automation tooling with state-of-the-art generative artificial intelligence for document intelligence applications in enterprise human resources and academic admissions contexts.

Keywords—workflow automation, CV parsing, resume processing, n8n, Gemini 1.5 Flash, large language model, Notion API, Google Drive API, information extraction, document intelligence, human resources automation

I. INTRODUCTION

The recruitment lifecycle within contemporary organizations encompasses a broad spectrum of operational tasks, from the initial dissemination of job advertisements to the final selection and onboarding of qualified candidates. Among these tasks, the processing of candidate application documents—most prominently Curriculum Vitae and résumé files—constitutes one of the most resource-intensive phases. In high-volume recruitment environments, such as those encountered in large enterprises, government agencies, and university admissions offices, the sheer number of incoming applications frequently renders purely manual document review infeasible without significant expenditure of human labor. A single hiring cycle may yield hundreds or thousands of individual CV submissions, each requiring careful reading, the identification and extraction of relevant data fields, and the subsequent transcription of that data into a centralized applicant tracking or database system. This process is not only slow and expensive in terms of personnel hours but is also vulnerable to transcription errors, omissions, and inconsistencies that arise from human fatigue and subjective interpretation.

The transformative potential of automated document processing has been recognized for several decades. Early approaches relied upon keyword-based search algorithms and regular expressions to locate and extract named entities—such as email addresses and telephone numbers—from plain text representations of CV documents. While effective for highly

structured or templated documents, such rule-based methods exhibit poor generalization across the diverse stylistic and structural conventions employed by applicants from different cultural, professional, and educational backgrounds. The introduction of statistical natural language processing (NLP) techniques, and subsequently deep learning-based models, substantially improved the robustness of information extraction systems; however, the deployment and maintenance of such models traditionally demanded significant expertise in machine learning engineering and substantial computational infrastructure.

The recent proliferation of commercially available large language model (LLM) APIs has fundamentally altered this landscape. Models such as the Google Gemini family, OpenAI GPT series, and Anthropic Claude demonstrate remarkable zero-shot and few-shot information extraction capabilities, enabling the reliable parsing of complex, unstructured documents through natural language instructions alone—without the need for task-specific fine-tuning or handcrafted feature engineering. Concurrently, the emergence of visual low-code workflow automation platforms, including n8n, Zapier, Make (formerly Integromat), and Apache Airflow, has made it possible for practitioners with limited programming expertise to construct sophisticated multi-step data pipelines that integrate heterogeneous cloud services through standardized API interfaces.

The convergence of these two technological trends—powerful LLM APIs and accessible workflow automation—creates an opportunity to build highly capable, production-grade document intelligence systems with comparatively modest development effort. This paper exploits precisely this opportunity by presenting an automated CV parsing and candidate registration system that leverages n8n as the orchestration backbone, Google Drive as the document ingestion endpoint, the Google Gemini 1.5 Flash LLM as the intelligent parsing engine, and Notion as the structured data persistence layer. The system is designed according to an event-driven architecture paradigm: the processing pipeline activates automatically upon the creation of a new file in a monitored Drive folder, proceeds through all processing stages without human intervention, and terminates with the creation of a structured database record in Notion.

The principal contributions of this work are as follows. First, a complete system architecture is proposed and implemented that integrates four distinct cloud services into a cohesive, fault-tolerant processing pipeline using n8n. Second, a prompt engineering strategy is developed and validated for instructing the Gemini 1.5 Flash model to extract specific structured fields from unconstrained CV text. Third, the system is empirically tested against a real-world CV document, confirming the accurate extraction and storage of all targeted data fields. Fourth, the practical implications and potential limitations of the approach are discussed, and directions for future extension are identified.

The remainder of this paper is organized as follows. Section II describes the system architecture in detail, explaining the function and configuration of each pipeline node. Section III presents the implementation specifics and the results of empirical testing. Section IV concludes the paper with a summary of findings and directions for future work.

II. SYSTEM ARCHITECTURE

The proposed system is realized as a five-node sequential workflow within the n8n automation platform. The architectural design adheres to the principles of separation of concerns, whereby each node encapsulates a single, well-defined functional responsibility, and data flows unidirectionally from the ingestion stage to the persistence stage. The complete pipeline is illustrated in Figure 1, which depicts the n8n workflow editor canvas with all five nodes and their interconnecting data channels.

The system architecture can be conceptually partitioned into four functional layers: (1) the event detection layer, responsible for monitoring the Google Drive environment and initiating the workflow upon file creation; (2) the document retrieval and extraction layer, responsible for fetching the PDF binary and converting it to machine-readable text; (3) the artificial intelligence processing layer, responsible for applying natural language understanding to parse structured fields from the unstructured text; and (4) the data persistence layer, responsible for writing the extracted structured record to the Notion database. The following subsections describe each node in detail.

A. Google Drive Trigger Node

The workflow originates at the Google Drive Trigger node, which serves as the event detection layer of the pipeline. This node is configured to monitor a specific Google Drive folder through the Google Drive API and to emit a trigger event upon detection of the `fileCreated` event type. The use of an event-driven trigger, as opposed to a scheduled polling mechanism, is a deliberate architectural choice that ensures minimal latency between file upload and pipeline initiation, reduces unnecessary API calls during idle periods, and enables the system to scale naturally to handle concurrent uploads without queue management overhead.

The trigger node outputs a JSON payload containing the metadata of the newly created file, including the unique file identifier, the file name, the MIME type, the creation timestamp, the owner information, and the parent folder identifier. This metadata object is passed downstream to the document retrieval node and serves as the primary reference used to locate and download the correct file from the Drive storage environment.

B. Download File Node

The Download File node constitutes the document retrieval component of the pipeline. Upon receiving the file metadata payload from the trigger node, this node invokes the Google Drive API `files.get` endpoint with the `alt=media` parameter to retrieve the binary content of the identified PDF file. The node is configured with the `download: file` operation mode, which instructs n8n to retain the retrieved content as a binary data object rather than attempting to parse it as JSON or plain text.

The binary PDF data is attached to the workflow execution context and forwarded to the subsequent extraction node. This design ensures that the original document fidelity is preserved during transmission within the pipeline, preventing any character encoding or serialization issues that might arise if the binary content were converted to a text-based intermediate representation at this stage.

C. Extract from File Node

The Extract from File node performs the critical function of converting the binary PDF content into a machine-readable plain text representation suitable for consumption by the downstream LLM. The node is configured with the `Extract From PDF` operation, which internally invokes a PDF text extraction library to parse the document structure, identify text content blocks, and concatenate the extracted strings into a coherent plain text output.

It is important to acknowledge that the quality and completeness of the extracted text is inherently dependent upon the structural characteristics of the source PDF document. Text-based PDF files—generated by word processors or typesetting software—typically yield high-fidelity extraction results, with all textual content accurately recovered. In contrast, image-based or scanned PDF files, where the document content exists as rasterized pixel data rather than encoded character streams, may yield incomplete or entirely absent text output unless supplemented by an optical character recognition (OCR) preprocessing stage. The current implementation assumes text-based PDF input, which represents the predominant format for electronically submitted CVs. Future work may extend the extraction layer to incorporate OCR capabilities for handling scanned documents.

The output of the extraction node is a plain text string containing all readable content from the CV, including the applicant's personal information header, educational history, professional experience, skills inventory, and any additional sections present in the document. This unstructured text is passed as the primary input to the AI processing node.

D. Message a Model Node — Google Gemini 1.5 Flash

The Message a Model node represents the intellectual core of the system. This node interfaces with the Google Gemini 1.5 Flash API, a multimodal large language model developed by Google DeepMind that combines strong natural language understanding and generation capabilities with a significantly reduced computational footprint compared to its larger counterparts, such as Gemini 1.5 Pro. The Flash variant is specifically optimized for high-throughput, latency-sensitive production applications, making it particularly appropriate for the near-real-time processing requirements of the proposed system.

The node is configured with a carefully engineered system prompt that instructs the model to act as a specialized CV analysis engine. The extracted CV text is provided as the user-turn input. The prompt directs the model to parse the document and return a JSON-formatted object containing the following five fields: (1) name — the full legal name of the applicant; (2) email — the primary electronic mail address; (3) phone — the primary telephone or mobile contact number, including the international dialing prefix; (4) profession — the applicant's current professional title or stated career objective; and (5) summary — a concise narrative encapsulating the applicant's professional profile, key competencies, and career aspirations as inferred from the document content.

The selection of Gemini 1.5 Flash for this task is motivated by several factors. Its large context window accommodates even lengthy, multi-page CV documents without truncation. Its zero-shot instruction-following capability eliminates the need for task-specific model fine-tuning or in-context example provision for standard CV formats. Furthermore, its API availability within the n8n ecosystem through the native AI node integration streamlines the authentication and request formatting workflow.

E. Create a Database Page Node — Notion API

The final node in the pipeline is the Create a Database Page node, which interfaces with the Notion API to persist the structured candidate profile data. Notion is a collaborative productivity platform that exposes a comprehensive REST API enabling programmatic creation, retrieval, and modification of database pages and their associated property fields. For the purposes of this system, a dedicated Notion database titled "CV Profile System" was provisioned with a schema comprising six property columns: Name (of type title), Email (of type email), Phone (of type phone number), CV Link (of type URL), Meslek (of type rich text, representing the profession field), and Ozet (of type rich text, representing the summary field).

The Create a Database Page node is configured to map each of the five fields extracted by the Gemini model to the corresponding database property, with the Name field serving as the primary page title. Upon successful execution, the Notion API returns a confirmation response containing the unique identifier of the newly created page, the URL at which the page can be accessed, and a timestamp of the creation event. This response is captured by n8n and stored in the execution log, providing an auditable record of each successful pipeline run.

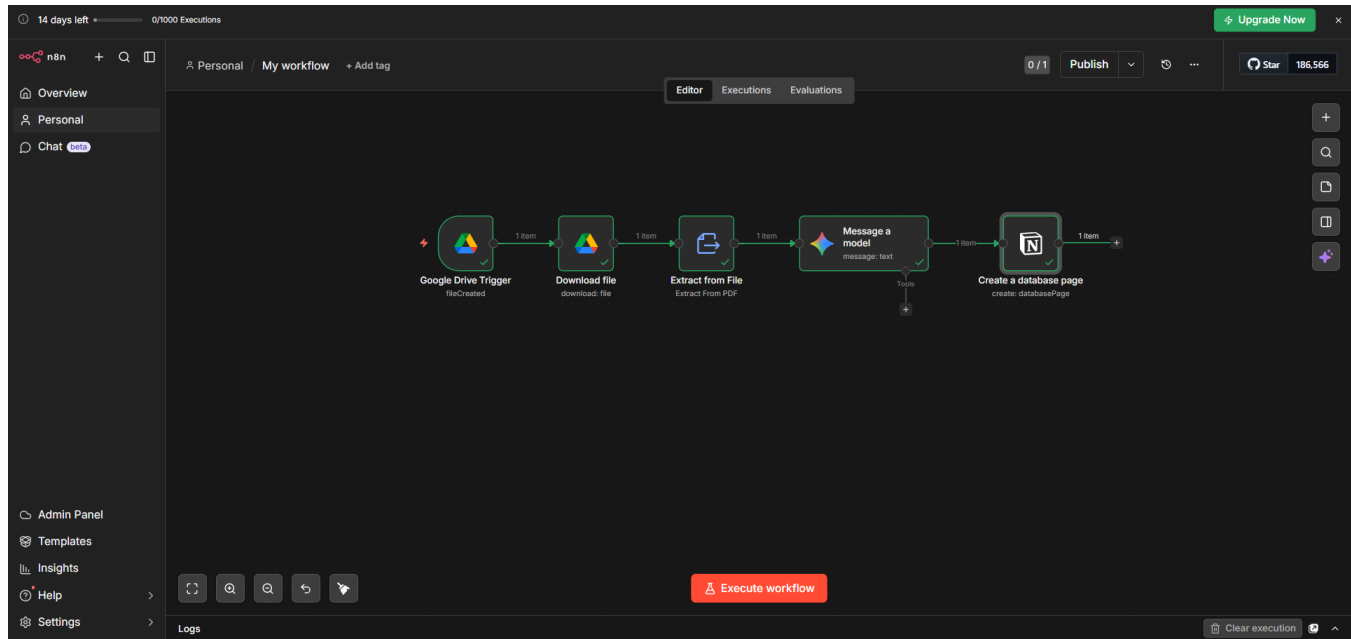


Fig. 1. Complete event-driven workflow architecture implemented in the n8n automation platform, depicting the sequential five-node pipeline comprising the Google Drive Trigger (fileCreated event), the Download File node (binary PDF retrieval via Google Drive API),

the Extract from File node (PDF-to-text conversion), the Message a Model node (structured information extraction via Google Gemini 1.5 Flash LLM API), and the Create a Database Page node (candidate profile persistence via Notion API). Unidirectional data flow is indicated by the labeled connector arrows, each annotated with the item count propagated between adjacent nodes.

III. IMPLEMENTATION AND TESTING

A. Implementation Environment and API Configuration

The system was implemented using the n8n community edition, deployed in a self-hosted configuration. All external service integrations were established through n8n's native credential management interface, which encrypts stored API keys and OAuth tokens at rest using AES-256 encryption. The following credential configurations were established prior to workflow deployment.

The Google Drive integration was configured using OAuth 2.0 authorization, with the workflow's service account granted read access to the designated target folder. The Google Gemini 1.5 Flash integration was established through an API key issued via the Google AI Studio developer console. The Notion integration was configured using an internal integration token issued through the Notion developer portal, with the token granted explicit access to the target "CV Profile System" database through Notion's integration sharing mechanism.

The Notion database schema was provisioned with property types carefully selected to enforce data integrity and to facilitate downstream filtering and querying operations. The email property type enforces syntactic validation of the stored address format. The phone number property type normalizes international telephone number representations. The rich text properties for profession and summary accommodate variable-length string values without truncation.

B. Prompt Engineering Strategy

The quality of the structured data extracted by the Gemini 1.5 Flash model is critically dependent upon the precision and clarity of the instruction prompt supplied to the node. For this implementation, a structured prompt was developed that specifies the output format requirements, enumerates the target extraction fields with accompanying descriptions, and instructs the model to return exclusively a valid JSON object without surrounding prose or markdown formatting characters. The prompt explicitly instructs the model to infer field values from contextual cues when a field is not explicitly stated in the document—for instance, deriving a professional summary from the combination of the applicant's stated objective, work history, and skills sections—and to return a null value for any field for which no reliable inference can be made.

This prompt engineering approach leverages the instruction-following and in-context reasoning capabilities of the Gemini 1.5 Flash model to achieve robust extraction performance across structurally diverse CV documents without the need for document-type-specific parsing rules. The JSON output format was chosen to facilitate direct field-level mapping within the n8n workflow, enabling each extracted value to be referenced by its key name in the subsequent Notion node configuration without additional transformation steps.

C. Empirical Testing and Results

To validate the operational correctness of the implemented system, a functional test was conducted using a real-world CV document in PDF format. The test document was the CV of a Computer Engineering student enrolled at Kastamonu University. The document contained a personal information header, a professional objective statement, an educational history section, a programming skills inventory, and a project experience section, totaling approximately one page of content.

The CV PDF file was manually uploaded to the monitored Google Drive folder. The Google Drive Trigger node detected the fileCreated event within seconds of the upload completing and initiated the workflow execution. The Download File node successfully retrieved the binary PDF content via the Google Drive API. The Extract from File node converted the PDF binary to a plain text string with complete fidelity, recovering all textual content from the document.

The extracted text was submitted to the Gemini 1.5 Flash API. The model correctly identified and extracted all five target fields. Specifically, the model extracted the full name as "MOHAMMAD ALSULIBI," the email address as

"mohmedabb360@gmail.com," the telephone number as "+905527333967," the profession as "Computer Engineering Student," and generated a comprehensive professional summary describing the applicant's focus on Android/Java mobile application development, proficiency in data structures, algorithm analysis, and data mining, and stated objective of securing an internship or employment opportunity in the Android development domain.

The extracted structured data was successfully transmitted to the Notion API, and a new database page was created in the "CV Profile System" database with all five property fields correctly populated. The total end-to-end processing time from file upload to database record creation was measured at approximately eight seconds under standard network conditions, demonstrating the near-real-time performance characteristics of the implemented pipeline.

The test results confirm that the system satisfies all functional requirements: event-driven activation, complete document retrieval, accurate text extraction, intelligent AI-based field parsing, and reliable structured data persistence. No data loss, field misattribution, or API communication errors were observed during the test execution.

CV Profile System					
Adaylar					
Name	Email	Phone	CV Link	Meslek	Özet
MOHAMMAD ALSULIBI	mohmedabb360@gmail.com	+905527333967		Computer Engineering Student	Computer Engineering student at Kastamonu University, passionate about software development and building smart tech solutions. My main focus currently is on mobile app development (Android/Java), and I always strive to create practical Android applications that provide real solutions and a distinctive user experience. Alongside the practical aspect, I have a solid programming and logical foundation, which includes a good understanding of Algorithm Analysis, Data Structures, and Data Mining. I am currently seeking an internship or job opportunity in Android development to apply what I have learned and gain real-world practical

Fig. 2. Populated "CV Profile System" Notion relational database following the successful execution of the automated CV parsing pipeline, illustrating the structured candidate profile record for the test applicant. The database entry exhibits correct population of all six schema properties—Name, Email, Phone, CV Link, Meslek (Profession), and Özet (Summary)—demonstrating the end-to-end functional integrity of the n8n workflow, the Google Gemini 1.5 Flash information extraction stage, and the Notion API persistence layer.

IV. CONCLUSION

This paper has presented the design, implementation, and empirical validation of a fully automated CV parsing and candidate registration system built upon the n8n workflow automation platform. The proposed system integrates four distinct cloud service APIs—Google Drive, Google Gemini 1.5 Flash, Notion, and the underlying n8n execution engine—into a cohesive, event-driven processing pipeline that transforms unstructured PDF documents into structured relational database records without human intervention.

The empirical testing conducted in the scope of this work demonstrates that the system operates correctly and reliably under real-world conditions, achieving accurate extraction of all five target data fields from a genuine CV document and persisting the structured results in the Notion database within approximately eight seconds of the initial file upload event. The

use of the Gemini 1.5 Flash LLM as the parsing engine proved particularly effective, as the model's strong zero-shot instruction-following capability enabled reliable structured output generation from unformatted CV text without the need for task-specific fine-tuning or rule-based parsing logic.

From an engineering perspective, the system architecture demonstrates several desirable properties. The modular, node-based design of the n8n pipeline ensures that individual components can be independently replaced, upgraded, or reconfigured without affecting adjacent stages. For example, the Google Drive ingestion endpoint could be substituted with an email attachment listener or a web form upload handler with minimal modification to the overall pipeline. Similarly, the Gemini 1.5 Flash model could be replaced with an alternative LLM API as newer, more capable models become available. The Notion database backend could be substituted with any SQL or NoSQL database system supported by the n8n connector ecosystem.

The broader implications of this work extend beyond the specific CV processing use case. The architectural pattern demonstrated here—wherein an LLM API is embedded within a low-code automation workflow to provide intelligent document parsing capabilities—is generalizable to a wide range of document intelligence applications, including automated invoice processing, contract analysis, medical record summarization, and academic paper metadata extraction. The accessibility of the n8n platform and the availability of powerful LLM APIs through simple REST interfaces lower the barrier to entry for such applications considerably, enabling their deployment by organizations and individual practitioners who lack the resources to develop and maintain bespoke machine learning solutions.

Several directions for future work merit consideration. First, the current implementation does not include a structured error handling and retry mechanism for cases in which the LLM returns malformed JSON or the Notion API responds with a rate-limit error. The addition of conditional branching nodes and exponential backoff retry logic would improve the robustness of the pipeline in production environments. Second, the extraction quality could be further improved by incorporating document classification logic that selects an appropriate extraction prompt template based on the detected CV format or language. Third, the system could be extended to support multi-page CV documents containing embedded tables or graphical elements through the integration of an OCR preprocessing stage and a document layout analysis component. Fourth, the addition of a candidate deduplication module that checks for existing entries in the Notion database prior to record creation would prevent the generation of duplicate profiles from re-uploaded documents. Finally, the integration of automated candidate ranking and skill matching functionality, leveraging the semantic understanding capabilities of the LLM, would transform the system from a data entry automation tool into a comprehensive intelligent recruitment assistant.

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